ASSIGMENT

PROGRAMMING USING C

WEEK – 04

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Week – 4(a)

Alice and Bob are playing a game called "Stone Game". Stone game is a two-player game. Let N be the total number of stones. In each turn, a player can remove either one stone or four stones. The player who picks the last stone, wins. They follow the "Ladies First" norm. Hence Alice is always the one to make the first move. Your task is to find out whether Alice can win, if both play the game optimally.

```
#include<stdio.h>
int main()
{
	int a,i=0,n,t;
	scanf("%d",&a);
	while(i<a)
	{
		scanf("%d",&n);
		t=n/4;
		if(t%2==0&&n%2==0)
		{
		printf("No\n");
		}
		else if(t%2==1&&n%2==1)
		{
		printf("No\n");
		}
		else
		{
		printf("Yes\n");
		}
	i++;
	}
}
```

	Input	Expected	Got	
✓	3 1 6 7	Yes Yes No	Yes Yes No	✓

Passed all tests! ✓

You are designing a poster which prints out numbers with a unique style applied to each of them. The styling is based on the number of closed paths or holes present in a given number.

```
#include<stdio.h>
int main()
{
      int a,b,n=0;
      scanf("%d",&a);
      while(a>0)
      {
            b=a%10;
            if(b==0||b==6||b==9||b==4)
            {
                  n=n+1;
            }
            else if(b==8)
            {
                  n=n+2;
            }
            a=a/10;
      }
      printf("%d",n);
}
```

	Input	Expected	Got	
✓	630	2	2	✓
✓	1288	4	4	✓

Passed all tests! ✓

The problem solvers have found a new Island for coding and named it as Philaland. These smart people were given a task to make a purchase of items at the Island easier by distributing various coins with different values. Manish has come up with a solution that if we make coins category starting from $1 till the maximum price of the item present on Island, then we can purchase any item easily. He added the following example to prove his point.

```
#include<stdio.h>
int main()
{
	int n,r=0;
	scanf("%d",&n);
	while(n!=0)
	{
		n=n/2;
		r=r+1;
	}
	printf("%d",r);
}
```

	Input	Expected	Got	
✓	10	4	4	✓
✓	5	3	3	✓
✓	20	5	5	✓
✓	500	9	9	✓
✓	1000	10	10	✓

Passed all tests! ✓

Week – 4(b)

A set of N numbers (separated by one space) is passed as input to the program. The program must identify the count of numbers where the number is odd number.

```
#include<stdio.h>
int main()
{
      int x=0,n;
      while(scanf("%d",&n)==1)
      {
            if(n%2!=0)
            {
                  x++;
            }
      }
      printf("%d",x);
      return 0;
}
```

	Input	Expected	Got	
✓	5 10 15 20 25 30 35 40 45 50	5	5	✓

Passed all tests! ✓

We can rotate digits by 180 degrees to form new digits. When 0, 1, 6, 8, 9 are rotated 180 degrees, they become 0, 1, 9, 8, 6 respectively. When 2, 3, 4, 5 and 7 are rotated 180 degrees, they become invalid. A *confusing number* is a number that when rotated 180 degrees becomes a **different** number with each digit valid.

```
#include<stdio.h>
int main()
{
	int x,n,y=1;
	scanf("%d",&n);
	while(n!=0&&y==1)
	{
		x=n%10;
		n=n/10;
		if(x==2||x==3||x==4||x==7)
		{
			y++;
		}
	}
	if(y==1)
	{
		printf("true");
	}
	else
	{
		printf("false");
	}
}
```

	Input	Expected	Got	
✓	6	true	true	✓
✓	89	true	true	✓
✓	25	false	false	✓

Passed all tests! ✓

A nutritionist is labeling all the best power foods in the market. Every food item arranged in a single line, will have a value beginning from 1 and increasing by 1 for each, until all items have a value associated with them. An item's value is the same as the number of macronutrients it has. For example, food item with value 1 has 1 macronutrient, food item with value 2 has 2 macronutrients, and incrementing in this fashion.

```
#include<stdio.h>
int main()
{
	long long n,t,i,nut=0;
	scanf("%lld%lld",&n,&t);
	for(i=1;i<=n;i++)
	{
		nut=nut+i;
		if(nut==t)
		{
			nut=nut-1;
		}
	}
	printf("%lld",nut%1000000007);
	return 0;
}
```

	Input	Expected	Got	
✓	2 2	3	3	✓
✓	2 1	2	2	✓
✓	3 3	5	5	✓

Passed all tests! ✓